

Math Thermo

class # 21

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Conservation of Total Energy

The total energy density is the sum of kinetic, potential, and internal energies:

$$(21.1) \quad \rho e = \rho \frac{1}{2} |\vec{v}|^2 + \rho \psi + \rho u.$$

Here e is the specific total energy density, ψ is the specific potential energy density and, u is the specific internal energy density.

For the internal energy density, let us assume a conservation equation of the form

$$(21.2) \quad \frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho u \vec{v}) = -\nabla \cdot \vec{J}_u^D + \sigma_u$$

where $\vec{v}$ is the barycentric velocity, $\vec{J}_u^D$ is the diffusive internal energy flux and σ_u is the source of internal energy.

For reasons that will be made clear later we will more commonly refer to $\vec{J}_u^D$ as the heat flux.

Previously, we argued that the internal energy was always conserved for an isolated system. But that assumption ignored kinetic and potential energy contributions. In the present case, energy may be transferred among the three types.

In an otherwise isolated system the total energy must be conserved. This means that the equation we produce must satisfy

$$(21.3) \quad \frac{\partial}{\partial t}(\rho e) + \nabla \cdot (\rho e \vec{v}) = -\nabla \cdot \vec{J}_E^D + \sigma_E; \quad \sigma_E \equiv 0.$$

In other words, the source term must be zero!

To obtain a specific form for (21.3) we add equations

$$(20.14) \quad \boxed{\begin{aligned} \frac{\partial}{\partial t} \left(\frac{1}{2} \rho |\vec{v}|^2 \right) + \nabla \cdot \left(\frac{1}{2} \rho |\vec{v}|^2 \vec{v} + \mathcal{J}^T \vec{v} \right) \\ = \mathcal{J} : \nabla \vec{v} + \rho \vec{F} \cdot \vec{v} \end{aligned}}$$

$$(20.15) \quad \boxed{\begin{aligned} \frac{\partial}{\partial t} (\rho \psi) + \nabla \cdot (\rho \psi \vec{v} + \sum_{i=1}^r \psi_i \vec{J}_i^D) \\ = -\rho \vec{F} \cdot \vec{v} + \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D. \end{aligned}}$$

and

$$(21.2) \quad \frac{\partial}{\partial t} (\rho u) + \nabla \cdot (\rho u \vec{v}) = -\nabla \cdot \vec{J}_U^D + \sigma_U,$$

obtaining

$$(21.4) \quad \vec{J}_E^D = \mathcal{J}^T \vec{v} + \sum_{i=1}^r \psi_i \vec{J}_i^D + \vec{J}_U^D$$

and

$$(21.5) \quad \sigma_E = \mathcal{J} : \nabla \vec{v} + \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D + \sigma_U.$$

In order that $\sigma_E = 0$, we must have

$$(21.6) \quad \sigma_v = -\mathcal{J} : \nabla \vec{v} - \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D$$

Therefore, the internal energy conservation equation must be

$$(21.7) \quad \frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho u \vec{v}) = -\nabla \cdot \vec{J}_v^D - \mathcal{J} : \nabla \vec{v} - \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D$$

Recapitulation Of Conservation Equations

Here are the equations we have thus far.

Total mass conservation:

$$(21.8) \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0.$$

Conservation of components $i \in \{1, \dots, r\}$:

$$(21.9) \quad \frac{\partial (\rho \phi_i)}{\partial t} + \nabla \cdot (\rho \phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D + \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k.$$

Conservation of total momentum:

$$(21.10) \quad \frac{\partial (\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \otimes \vec{v} + \mathcal{J}) = -\sum_{i=1}^r \rho_i \nabla \psi_i.$$

And, finally, conservation of internal energy:

(21.11)

$$\frac{\partial}{\partial t}(\rho u) + \nabla \cdot (\rho u \vec{v}) = -\nabla \cdot \vec{J}_0^D - \mathcal{J} : \nabla \vec{v} - \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D.$$

let us recall some constraints in our system.

(21.12)

$$\sum_{i=1}^r \phi_i = 1, \quad (\phi_i = \frac{\rho_i}{\rho})$$

(21.13)

$$\mathcal{J}^T = \mathcal{J},$$

(21.14)

$$\sum_{i=1}^r \vec{J}_i^D = \vec{0},$$

and

(21.15)

$$\sum_{i=1}^r \gamma_{i,k} = 0, \quad 1 \leq k \leq K.$$

The unknowns in the system are the component fluxes,

$$\vec{J}_i^D, \quad i = 2, \dots, r,$$

the structure of the Cauchy stress tensor

$$\mathcal{J}^T = \mathcal{J}$$

and the heat flux,

$$\vec{J}_0^D$$

and the reaction rates

$$\tilde{\sigma}_k, \quad k = 1, \dots, K.$$

These must be determined by constitutive relations designed to guarantee that the entropy increases.

A Gibbs Relation

Up to now, we have been writing

$$\tilde{S} = \tilde{S}(U, V, N_1, \dots, N_r)$$

for the entropy of an r -component mixture. Here N_i is the number of moles of component i in the mixture. We could equally well write

$$\tilde{S} = \tilde{S}(U, V, M_1, \dots, M_r)$$

where M_i is the mass of component i , we only need a conversion factor:

$$M_i = X_i N_i, \quad i=1, \dots, r,$$

where X_i is mass per mole of species i . This is of course a constant.

In this case the Gibbs relation is similar to what we had before:

$$(21.16) \quad d\tilde{S} = \frac{1}{T_s} dU + \frac{P_s}{T_s} dV - \sum_{i=1}^r \frac{\mu_{s,i}}{T_s} dM_i$$

where in a slight abuse of notation

$$\mu_{s,i} = -T_s \frac{\partial \tilde{S}}{\partial M_i}$$

is our modified chemical potential.

We recall that $\tilde{S}$ is a homogeneous of degree

one function. let M be the total mass and set

$$u := \frac{U}{M}, \text{ the specific energy}$$

$$\rho := \frac{M}{V}, \text{ mass density}$$

$$v := \frac{1}{\rho} = \frac{V}{M}, \text{ the specific volume}$$

$$\phi_i := \frac{M_i}{M}, \text{ } i^{\text{th}} \text{ mass fraction}$$

$$\tilde{s} := \frac{\tilde{S}}{M}, \text{ the specific entropy}$$

Then,

$$\begin{aligned}\tilde{s} &= \tilde{S}\left(\frac{U}{M}, \frac{V}{M}, \frac{M_1}{M}, \dots, \frac{M_r}{M}\right) \\ &= \tilde{S}(u, v, \phi_1, \dots, \phi_r)\end{aligned}$$

It is easy to show that

$$\frac{\partial \tilde{S}}{\partial u} = \frac{1}{T_s}$$

$$\frac{\partial \tilde{S}}{\partial v} = \frac{P_s}{T_s}$$

$$\frac{\partial \tilde{S}}{\partial \phi_i} = -\frac{\mu_i}{T_s}$$

Thus,

$$(21.17) \quad d\tilde{s} = \frac{1}{T} du + \frac{P}{T} dv - \sum_{i=1}^r \frac{\mu_i}{T} d\phi_i$$

is the Gibbs relation for the specific entropy.

Definition (21.1) We say that a fluid mixture that has not achieved equilibrium satisfies local thermodynamic equilibrium iff pointwise in the system and for all time the Gibbs relation is valid

$$(21.17) \quad ds = \frac{1}{T} du + \frac{P}{T} dv - \sum_{i=1}^r \frac{\mu_i}{T} d\phi_i,$$

where u , v , and ϕ_i are smooth functions of space and time.

Postulate I : (of Non-Equilibrium Thermodynamics) For a fluid mixture that satisfies local thermodynamic equilibrium, with mass-averaged velocity $\vec{v}$, it holds at every point in time and space that

$$(21.18) \quad T \frac{D \circ s}{Dt} = \frac{D \circ u}{Dt} + P \frac{D \circ v}{Dt} - \sum_{i=1}^r \mu_i \frac{D \circ \phi_i}{Dt},$$

where, recall,

$$\frac{D \circ q}{Dt} = \frac{\partial q}{\partial t} + \vec{v} \cdot \nabla q$$

is the material derivative with respect to the mass-averaged velocity.

The Specific Volume

Recall the continuity equation:

$$(21.19) \quad \partial_t \rho + \nabla \cdot (\rho \vec{v}) = 0.$$

Now observe that

$$\begin{aligned}
 \partial_t(1/\rho) &= -1/\rho^2 \partial_t \rho \\
 &\stackrel{(21.19)}{=} -1/\rho^2 (-\nabla \cdot (\rho \vec{v})) \\
 &= \frac{1}{\rho^2} \nabla \cdot (\rho \vec{v}) \\
 &= \frac{1}{\rho^2} \rho \nabla \cdot \vec{v} + \vec{v} \cdot \frac{1}{\rho^2} \nabla \rho \\
 &= \frac{1}{\rho} \nabla \cdot \vec{v} - \vec{v} \cdot \nabla(1/\rho)
 \end{aligned}$$

Thus,

$$\partial_t(1/\rho) + \vec{v} \cdot \nabla(1/\rho) = \frac{1}{\rho} \nabla \cdot \vec{v}$$

In other words, since $v = 1/\rho$,

$$\partial_t v + \vec{v} \cdot \nabla v = v \nabla \cdot \vec{v}.$$

Equivalently,

$$(21.20) \quad \frac{D\vec{v} \cdot \vec{v}}{Dt} = v \nabla \cdot \vec{v}$$

and

$$(21.21) \quad \rho \frac{D\vec{v}}{Dt} = \nabla \cdot \vec{\tau}$$

From (21.18) we trivially have

$$(21.22) \quad \rho T \frac{Ds}{Dt} = \rho \frac{D\vec{v} \cdot \vec{u}}{Dt} + \rho P \frac{D\vec{v} \cdot \vec{v}}{Dt} - \sum_{i=1}^r \mu_i \rho \frac{D\vec{v} \cdot \vec{\phi}_i}{Dt}.$$

Now, we use (21.21) and the conservation equations (21.9) and (21.11),

$$(21.9) \quad \frac{\partial(\rho \phi_i)}{\partial t} + \nabla \cdot (\rho \phi_i \vec{u}) = -\nabla \cdot \vec{J}_i^D + \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k,$$

$$(21.11) \quad \frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \vec{v}) = -\nabla \cdot \vec{J}_0^D \\ - \mathcal{J} : \nabla \vec{v} - \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D,$$

recalling that

$$\rho \frac{D \vec{u}}{Dt} = \frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \vec{v}),$$

et cetera, to obtain

$$\rho T \frac{D \vec{s}}{Dt} = -\nabla \cdot \vec{J}_0^D - \mathcal{J} : \nabla \vec{v} - \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D \\ + P \nabla \cdot \vec{v} \\ - \sum_{i=1}^r \mu_i \left\{ -\nabla \cdot \vec{J}_i^D + \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k \right\}$$

Therefore,

$$\rho \frac{D \vec{s}}{Dt} = -\frac{\nabla \cdot \vec{J}_0^D}{T} - \frac{\mathcal{J} : \nabla \vec{v}}{T} - \frac{1}{T} \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D \\ + \frac{P \nabla \cdot \vec{v}}{T} \\ - \sum_{i=1}^r \frac{\mu_i}{T} \left\{ -\nabla \cdot \vec{J}_i^D + \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k \right\}$$

Now observe that

$$\frac{1}{T} \nabla \cdot \vec{J}_0^D = \nabla \cdot \left(\frac{\vec{J}_0^D}{T} \right) - \vec{J}_0^D \cdot \nabla \left(\frac{1}{T} \right),$$

and

$$\frac{\mu_i}{T} \nabla \cdot \vec{J}_i^D = \nabla \cdot \left(\frac{\mu_i}{T} \vec{J}_i^D \right) - \vec{J}_i^D \cdot \nabla \left(\frac{\mu_i}{T} \right).$$

Then,

(21.23)

$$\begin{aligned} \rho \frac{D_{\vec{v}} S}{Dt} = & -\nabla \cdot \left(\frac{\vec{J}_v^D}{T} \right) + \vec{J}_v^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\mathcal{J} : \nabla \vec{v}}{T} \\ & + \frac{P \nabla \cdot \vec{v}}{T} - \frac{1}{T} \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D \\ & + \sum_{i=1}^r \left\{ \nabla \cdot \left(\frac{\mu_i}{T} \vec{J}_i^D \right) - \vec{J}_i^D \cdot \nabla \left(\frac{\mu_i}{T} \right) \right\} \\ & - \sum_{i=1}^r \frac{\mu_i}{T} \sum_{k=1}^K \gamma_{i,k} \vec{F}_k \end{aligned}$$

Now, we need to specify the structure of the Cauchy stress tensor. Typically, $\mathcal{J}$ is decomposed into the sum of a hydrostatic part and a symmetric, viscous tensor part,

$$\mathcal{J} = P \mathbf{I}_3 + \Pi,$$

where P is the scalar pressure found in the Gibbs relation (21.18) and

$$\Pi \in \mathbb{R}_{\text{sym}}^{3 \times 3}.$$

This decomposition implies that

$$\begin{aligned} \frac{\mathcal{J} : \nabla \vec{v}}{T} &= \frac{1}{T} \{ P \mathbf{I}_3 : \nabla \vec{v} + \Pi : \nabla \vec{v} \} \\ &= \frac{1}{T} P \nabla \cdot \vec{v} + \frac{1}{T} \Pi : \nabla \vec{v} \end{aligned}$$

Expression (21.23) becomes

$$\begin{aligned}
 (21.24) \quad \rho \frac{D_{\vec{v}} S}{Dt} = & -\nabla \cdot \left(\frac{\vec{J}_v^D}{T} - \sum_{i=1}^r \frac{\mu_i}{T} \vec{J}_i^D \right) \\
 & + \vec{J}_v^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\Pi : \nabla \vec{v}}{T} \\
 & - \frac{1}{T} \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D - \sum_{i=1}^r \vec{J}_i^D \cdot \nabla \left(\frac{\mu_i}{T} \right) \\
 & - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\sigma}_k
 \end{aligned}$$

where A_k is the k^{th} chemical affinity,

$$A_k := \sum_{i=1}^r \nu_{i,k} \mu_i, \quad k=1, \dots, K.$$

Therefore, we may write

$$(21.25) \quad \rho \frac{Ds}{Dt} = -\nabla \cdot \vec{J}_s^D + \sigma_s$$

where

$$(21.26) \quad \vec{J}_s^D := \frac{1}{T} \left(\vec{J}_v^D - \sum_{i=1}^r \mu_i \vec{J}_i^D \right)$$

and

$$\begin{aligned}
 (21.27) \quad \sigma_s = & \vec{J}_v^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\Pi : \nabla \vec{v}}{T} \\
 & - \frac{1}{T} \sum_{i=1}^r \vec{J}_i^D \cdot \left(T \nabla \left(\frac{\mu_i}{T} \right) + \nabla \psi_i \right) \\
 & - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\sigma}_k.
 \end{aligned}$$

Postulate II : (of Non-Equilibrium Thermodynamics) For a fluid mixture that satisfies local thermodynamic equilibrium the specific entropy satisfies the conservation equation

$$\rho \frac{Ds}{Dt} = - \nabla \cdot \vec{J}_s^D + \sigma_s$$

where

$$\sigma_s \geq 0$$

for all points in space and time.